Higher National Diploma in Information Technology
First year, second Semester Examination – 2023
IT2012 Fundamentals of Programming

Instructions for Candidates:
Answer only five questions

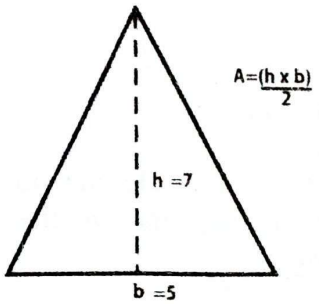
No. of questions: 6
No. of pages: 2
Time: 3 hours

Question 01

I. What is a translator software (02 marks)
II. Compare the differences between Interpreter and compiler (04 marks)
III. Explain the following Java escape sequences (06 marks)
   a. \\
   b. \b
   c. \"
IV. Briefly explain followings (08 marks)
   a. Source code
   b. Object code
   c. Machine code
   d. Virtual machine

Question 02

I. What is meant by variable in Java ? (02 marks)
II. What is the different between keyword and identifier? (04 marks)
III. Mention the use of following Java data types, and size of them in bits (06 marks)
   a. Char
   b. boolean
   c. int
IV. Consider the following image of a triangle, and write the Java code to do the followings (08 marks)



   a. Declare int type variables h for height and b for base and initialize values.
   b. Declare the float type variable A for area, and calculate the area.

Question 03

I. What is meant by expression in computer programming? (02 marks)

II. Mention the differences between operator and operand (04 marks)

III. Assume the price of an item is denoted in 'P', the quantity is denoted in 'Q', and the discount percentage is denoted in 'D', and the service charge is 'C'. Write a Java Program to Calculate the Total Bill (10 marks)

IV. Consider the following Java code segment and write the output (1*4=04 marks)

```
int a = 7;
int b = 5;
System.out.println(a > b && b > 3);
System.out.println(a < b || b == 5);
System.out.println(!(a == b));
System.out.println(a >= 7 && b != 5);
```

Question 04

I. What is meant by control structure in computer programming? (02 marks)

II. Compare the differences between while and do-while (04 marks)

III. Write a Java program using a while loop to display the following number sequence: 3, 6, 9, 12, 15. (06 marks)

IV. A company pays Rs.2500 per day for its employees. In addition, employees receive Rs.300 for 1 hour of overtime (OT) and Rs.700 for 2 hours of OT. Write a Java program with switch-case to input the number of overtime hours (0, 1, or 2) and calculate the total pay for the day. (08 marks)

Question 05

I. Write the Java code for printing the all elements of an array named Marks, using while loop and length property. (04 marks)

II. Explain the use of following blocks in exception handling (04 marks)

a. try block

b. catch block

III. Write a Java program to define a class named Employee with three data members: empID (an integer), empName (a string), and address (a string). The class should include a method named addEmployee() to assign values to these data members and a method named displayInfo() to print the employee's details.

Demonstrate the use of this class by creating an object, invoking the addEmployee() method to set values, and then calling displayInfo() to output the employee's information.. (12 marks)

Question 06

I. What is meant Graphical User Interface in a computer program (02 marks)

II. What are the advantages of GUI comparing to CLI. (04 marks)

III. Assume you need to develop a Java Application for whole sale coconut oil store, to input the price of one liter and number of liters customer buy, and find the customer bill value. Write the necessary Java coding for following (06 marks)

a. Declare JTextField named txtPrice

b. Declare JTextField named txtQty

c. Declare JTextField named txtValue

d. Initialize txtPrice

e. Initialize txtQty

f. Initialize txtValue

IV. Write the necessary Java coding for calculate the bill value and display the answer in label named lblResult. (8 marks)